# Concept Map (Conceptual Model), Version 4.0

Ontology-Guided Agentic Capabilities Orchestration (AC-001)

Team AgenticCapabilities · Ontology Engineering 2026, Assignment 10 · Version 4.0, November 6, 2026

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This model covers the orchestration layer of use case AC-001: what a user request asks for, which capabilities, skills and tools can meet it, how candidate orchestration plans are built and compared, and what the knowledge graph (KG) records when a plan runs. Version 4.0 follows the Assignment 10 ontology, OE_10_TeamAgenticCapabilities.rdf: Figures 9a to 9e draw the shortcut properties added so that the SPARQL queries can follow one property instead of a path, and Figure 7a adds confirmsPrecondition. Every class, is-a line, labeled arrow, multiplicity and attribute in Figures 1a to 9e is a class, subclass axiom, object property, cardinality restriction or datatype property of the ontology, with the same name; a script checks all 356 of them against the ontology file, and a second script checks that all 133 classes and properties declared in the ontology file, reused ones included, appear in a figure. The ontology has 64 classes, 49 object properties and 8 datatype properties, and its 162 test individuals are in OE_10_TeamAgenticCapabilities_Individuals.rdf. Its deepest chains have three levels below a reused top class, for example RejectedOrchestrationPlan, CandidateOrchestrationPlan, OrchestrationPlan under pplan:Plan. The three competency questions the ontology is scoped to, CQ 1, CQ 3 and CQ 5, are marked in the table at the end; every competency question has a query in OE_10_TeamAgenticCapabilities_Queries.txt.

**Notation.** UML 2.5.1 class diagrams drawn with PlantUML. A box is a class; attributes inside a box are datatype properties. A line with a hollow triangle points from a subclass to its superclass (is a). A labeled arrow is an object property drawn from its domain to its range; a dashed arrow whose name starts with "/" is derived, meaning the reasoner infers it. «AgentO», «P-Plan», «PROV-O», «OWL-S», «Commons» and «BEAM» mark classes reused from those vocabularies; «defined» marks classes whose members the reasoner infers from a definition, and an «enumeration» box lists the individuals that are the members of a class. A gray box is a class detailed in another figure. Property prefixes name reused vocabularies: agento: AgentO [3], prov: PROV-O [9], process: OWL-S [8], pplan: P-Plan [12]; properties without a prefix belong to the team ontology. In an example, a box written name : Class is an individual and an orange dashed arrow is a relation the reasoner infers. A number at the end of an arrow is a multiplicity from a cardinality restriction: 1 means exactly one, 1..* at least one, 0..1 at most one. An attribute marked [0..1] is a functional property. A note {complete, disjoint} under a class says that each of its members belongs to exactly one of the subclasses drawn below it ({complete, overlapping}: to at least one). A loop labeled isX on a class X is a helper property that links each member of X to itself (X EquivalentTo isX Self); property chains use it to require a class, which a chain cannot name directly. Notes and the tables under some figures give axioms in Protégé's Manchester syntax; the tables under Figures 9a to 9e give the property chains, written chain SubPropertyOf shortcut.

**Changes in Version 4.0 (Assignment 10).** What is new since Version 3.0 is blue: new arrows and labels in the diagrams, new figure captions, rewritten paragraphs, new table rows, and a "Changes in 4.0" paragraph under each figure that changed. The changes:

- Figures 9a to 9e (new) draw 21 of the 22 object properties added to the ontology for Assignment 10: 14 shortcuts defined by property chains, from a request, a skill, a tool, a workflow step and a plan; hasUserConstraint, which hasHardConstraint and hasSoftConstraint specialize; and the six helper properties isRequest, isPreferenceSetting, isSkill, isTool, isHardConstraint and isSoftConstraint that the chains use. The table under each figure gives the chains.
- Figure 7a: the 22nd new property, confirmsPrecondition, links an Invocation to the Precondition it checked (CQ 9).
- Figure 8c and references [15] to [17], which described terms outside the ontology, are removed; the Assignment 9 copy had already left out the figure.
- The competency-question paragraphs and the table at the end now describe the queries of OE_10_TeamAgenticCapabilities_Queries.txt. The paragraphs under Figures 1a, 1b and 5a, which were empty in the Assignment 9 copy, are written again.

- The diagrams of Figures 1a to 6b, 7b, 8a and 8b and their axiom tables are unchanged.

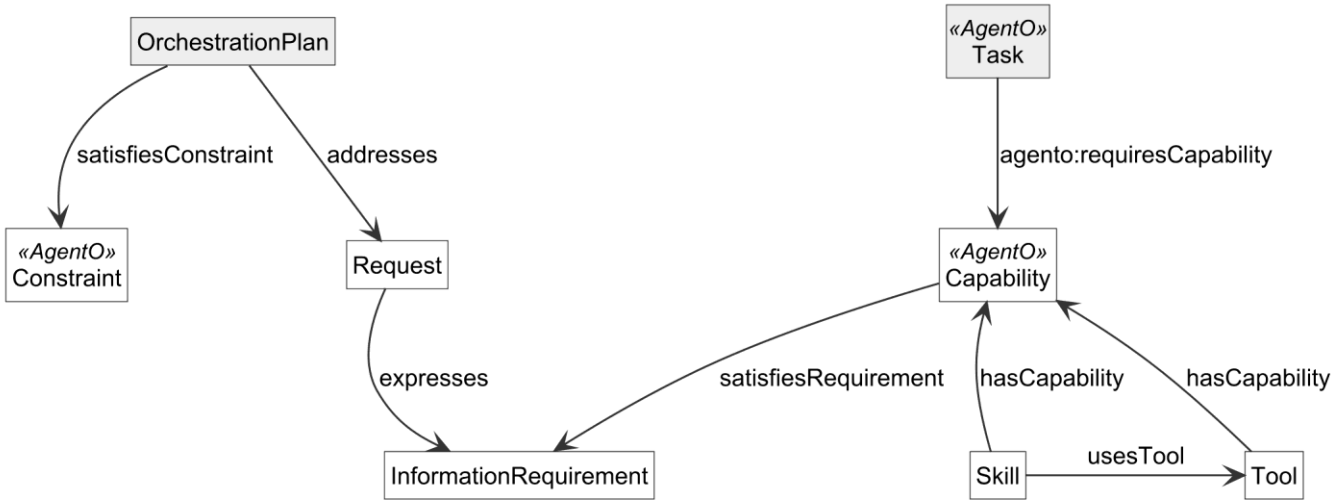



**Figure 1a. Overview, part 1: requests, requirements and capabilities**

**Shows.** What a request asks for and what can provide it. A Request expresses InformationRequirements, which Capabilities satisfy; Skills and Tools have Capabilities, and a Skill uses Tools. An OrchestrationPlan addresses the Request and satisfies Constraints, and each of its Tasks requires a Capability. The two gray classes are the links to Figure 1b.

**Competency questions.** CQ 3 (Jessica, a focus question) follows Request → InformationRequirement ← Capability ← Skill with stated links only. CQ 1 step 1 (Ken, a focus question) asks the same for Ken's request through the shortcuts of Figures 9a and 9b, which link a request to its hard constraints and a skill or tool to the requirements and hard constraints it meets. CQ 5 (Dante, a focus question) follows Task → Capability → InformationRequirement for the tasks of the selected plan (Figure 1b).

**Changes in 4.0.** Competency-question paragraph rewritten for the Assignment 10 queries (it was empty in the Assignment 9 copy).

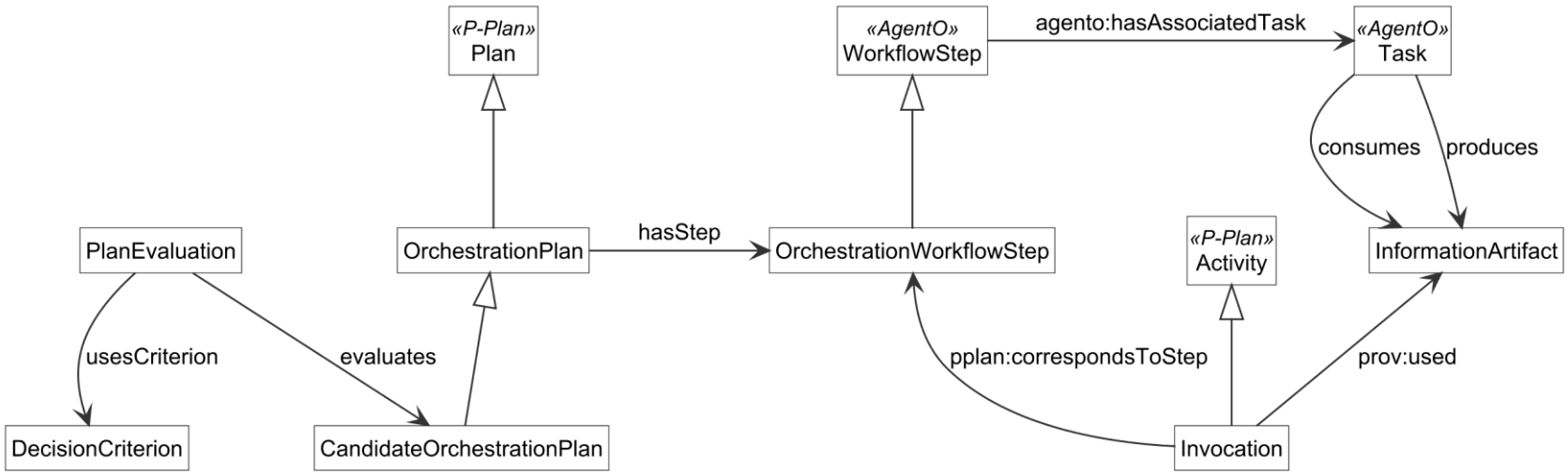



**Figure 1b. Overview, part 2: plans, steps and execution**

**Shows.** How a plan is built, chosen and run. An OrchestrationPlan is a P-Plan Plan; its OrchestrationWorkflowSteps (hasStep), a kind of AgentO WorkflowStep, are each associated with a Task, and a Task consumes and produces InformationArtifacts. PlanEvaluations evaluate CandidateOrchestrationPlans on DecisionCriteria. When the plan runs, each Invocation, a P-Plan Activity, corresponds to one OrchestrationWorkflowStep and uses InformationArtifacts.

**Competency questions.** CQ 1 (Ken, a focus question): query 1 compares his CandidateOrchestrationPlans through their PlanEvaluations and the inferred selected or rejected class (Figure 6a), and query 2 lists the selected plan's OrchestrationWorkflowSteps (hasStep) with the Task of each (agento:hasAssociatedTask). CQ 5 (Dante, a focus question) follows OrchestrationPlan → OrchestrationWorkflowStep → Task here and continues in Figure 1a. CQ 9 reads the Invocations that an ExecutionTrace records (Figures 7a and 7b).

**Changes in 4.0.** Competency-question paragraph rewritten for the Assignment 10 queries (it was empty in the Assignment 9 copy).

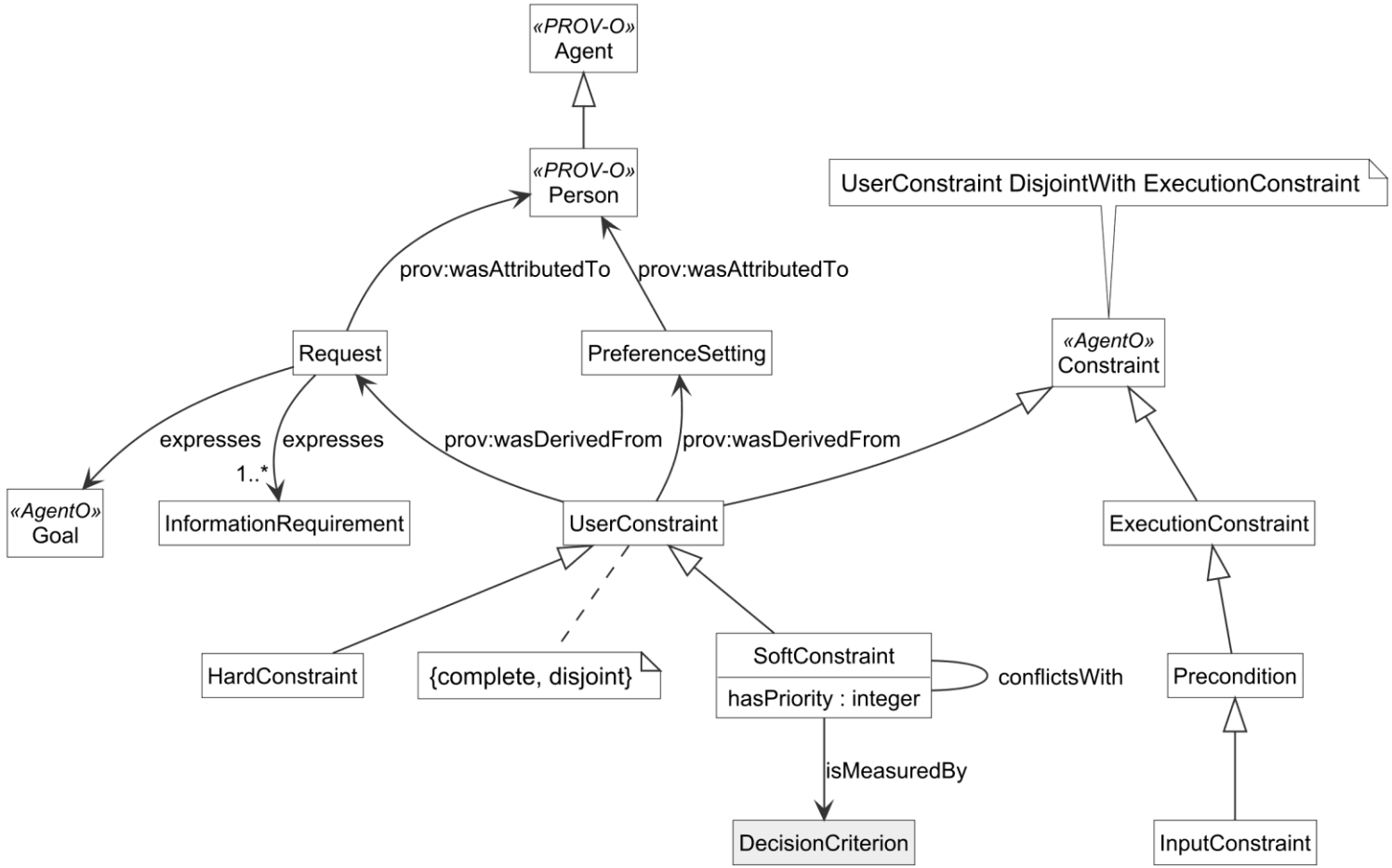



**Figure 2. Requests, requirements and constraints**

**Shows.** What the NLP tool and harness extract in basic flow steps 2–3: the Goals and InformationRequirements a Request expresses, and the UserConstraints that steer plan selection. prov:wasDerivedFrom records whether a constraint came from the Request or from the user's PreferenceSettings. PreferenceSettings are what a user saves before making requests; no step consumes or produces them, so they are not information artifacts and stay outside the artifact hierarchy of Figure 8a. A Request and a PreferenceSetting are both attributed to the prov:Person who made them (prov:wasAttributedTo), which links a request to the settings of the same user.

**Modeling decisions.** User constraints are split by how they are used. A HardConstraint must hold, for example Jessica's safety screening, which her PreferenceSettings turn on for her prescription medication; a plan that violates it is rejected. A SoftConstraint is something to optimize, for example Ken's "minimize external tool calls". Each SoftConstraint has a user-given priority (hasPriority) and is measured by a DecisionCriterion (Figure 6a). conflictsWith links two soft constraints that push the same criterion in opposite directions, such as maximizing and minimizing tool use (alternate flow 3);

the agent then asks the user to revise them. Constraints that only trade off, like Olivia's strong evidence and few tool calls, are not conflicts: their priorities weight them. ExecutionConstraints come from planning: a Precondition must hold before a tool or skill runs, and an InputConstraint is a precondition on one input value. A constraint comes either from the user or from planning, so UserConstraint and ExecutionConstraint are disjoint. Every UserConstraint is a HardConstraint or a SoftConstraint and never both ({complete, disjoint}), so a constraint stored only as a user constraint still has one of the two kinds, and one stated as both makes the KG inconsistent. A Request expresses at least one InformationRequirement (1..*). The axioms:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `UserConstraint EquivalentTo HardConstraint or SoftConstraint` | every user constraint is hard or soft |
| `HardConstraint DisjointWith SoftConstraint` | no constraint is both |
| `UserConstraint DisjointWith ExecutionConstraint` | a constraint comes from the user or from planning |
| `Request SubClassOf expresses min 1 InformationRequirement` | a request asks for at least one piece of information |

**Competency questions.** The CQ 3 query follows these links as stated: Jessica's safety-screening HardConstraint is derived from her PreferenceSetting, which is attributed to her as a prov:Person, as her request is. The CQ 1, CQ 2, CQ 4 and CQ 7 queries reach a request's constraints through the shortcuts hasHardConstraint and hasSoftConstraint (Figure 9a), which the reasoner infers from the same links. Alternate flow 3 uses conflictsWith.

**Changes in 4.0.** Competency-question paragraph updated: the queries now use the request shortcuts of Figure 9a.

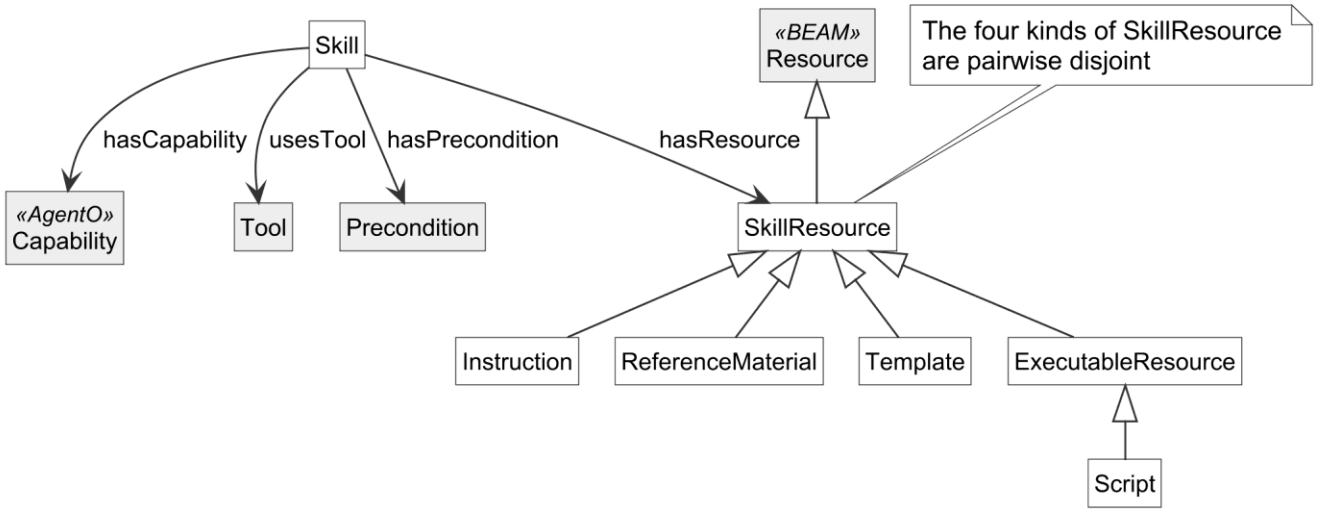



**Figure 3a. Skills and skill resources**

**Shows.** A Skill has Capabilities (Figure 3b), bundles SkillResources, uses Tools and has Preconditions. The resources are Instructions, ReferenceMaterial, Templates and ExecutableResources such as Scripts, as in the skill packages of SkillsBench [5].

**Modeling decisions.** SkillResource is the team's class for the parts of a skill package. It sits under BEAM's Resource, the class AgentO uses for what tasks require and produce [3], and its four kinds are disjoint. A skill's precondition uses the team's hasPrecondition: OWL-S's process:hasPrecondition has the domain process:Process and would make every skill a process. Tools keep process:hasPrecondition (Figure 4).

**Competency questions.** CQ 9 checks a skill's preconditions before it runs; CQ 3 and CQ 5 start from the skills' capabilities in Figure 3b.

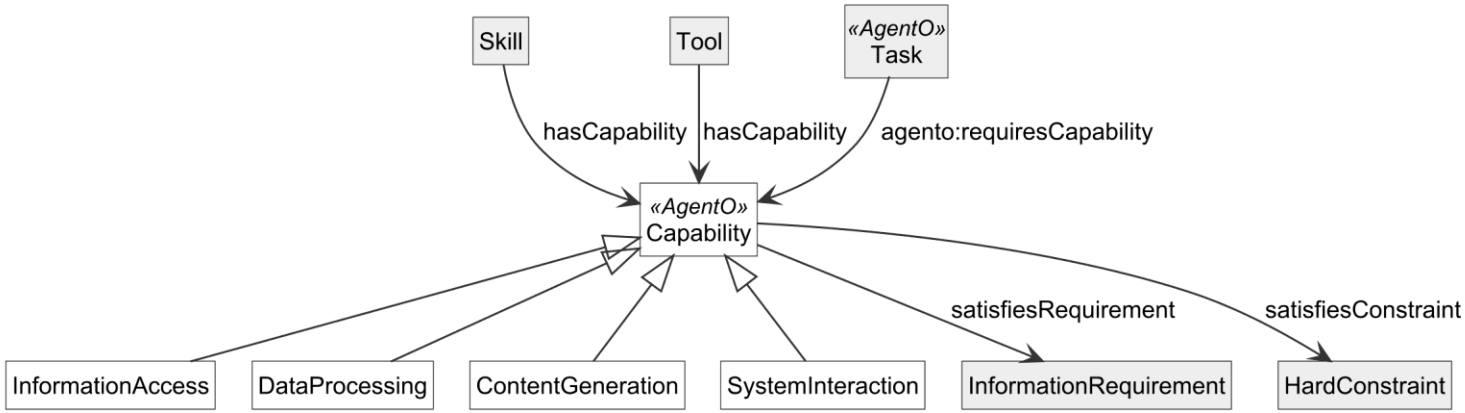



**Figure 3b. Capabilities**

**Shows.** Capabilities come in four kinds, following Mulang et al. [1]: InformationAccess, DataProcessing, ContentGeneration and SystemInteraction. Skills and Tools have Capabilities and Tasks require them. A Capability satisfies InformationRequirements and HardConstraints.

**Modeling decisions.** Skill is not a subclass of Capability: a skill is a packaged specification that has capabilities, and the same capability, such as screening for interactions, can come from a skill or directly from a tool. Keeping Capability separate lets the reasoner compare the two. hasCapability widens AgentO's property of the same name, whose domain is only Tool. The team's Capability is the same class as agento:Capability (equivalent classes).

**Competency questions.** CQ 3 (Jessica, a focus question): her request expresses three information requirements, and her preference setting yields the safety-screening HardConstraint; the capabilities that satisfy them belong to the personalized supplement recommendation, evidence verification, product selection and supplement safety screening skills, which is the answer the reasoner returns on the example data. CQ 5 (Dante, a focus question) reads the same links from the tasks of his plan: each task requires a capability, and that capability satisfies one of his information requirements or his safety-screening hard constraint. CQ 1 step 1 (Ken) reads the same links through the skill and tool shortcuts of Figure 9b.

**Changes in 4.0.** Last sentence added (Figure 9b).

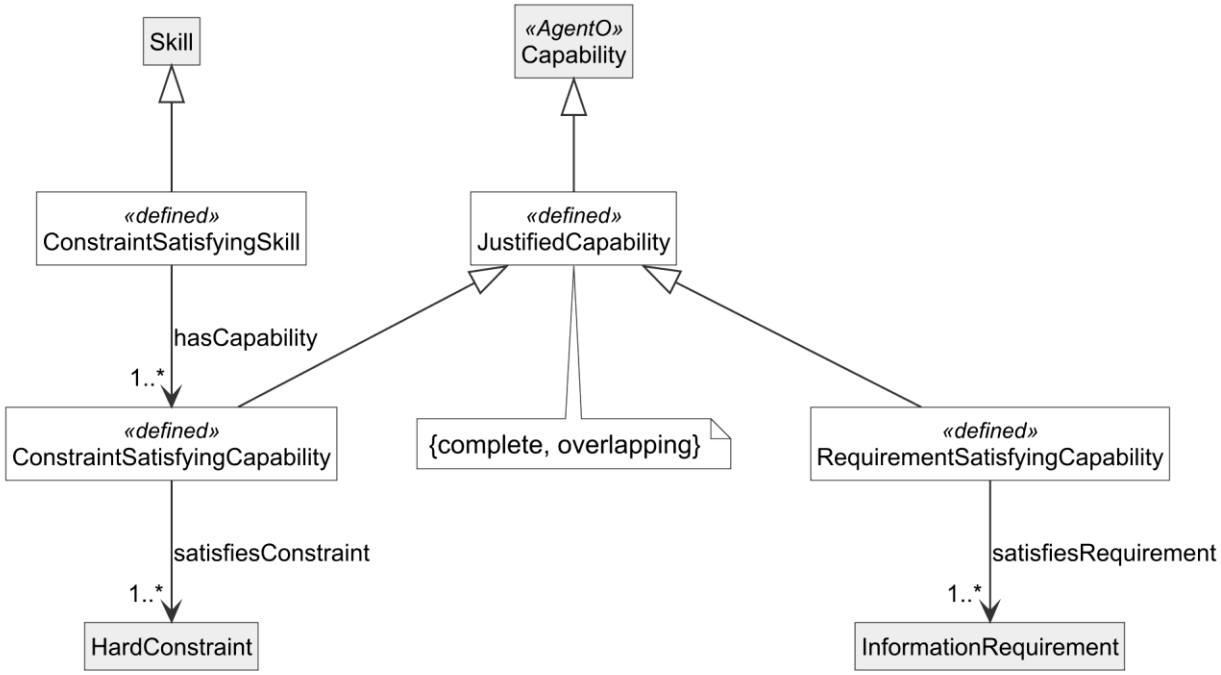



**Figure 3c. Justified capabilities and constraint-satisfying skills**

**Shows.** Four «defined» classes whose members the reasoner infers from the links of Figures 3b and 2. A RequirementSatisfyingCapability satisfies at least one InformationRequirement, a ConstraintSatisfyingCapability satisfies at least one HardConstraint, and a JustifiedCapability is either kind or both ({complete, overlapping}). A ConstraintSatisfyingSkill has at least one ConstraintSatisfyingCapability. The definitions:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `RequirementSatisfyingCapability EquivalentTo Capability and (satisfiesRequirement some InformationRequirement)` | capability that meets a requirement |
| `ConstraintSatisfyingCapability EquivalentTo Capability and (satisfiesConstraint some HardConstraint)` | capability that meets a hard constraint |
| `JustifiedCapability EquivalentTo RequirementSatisfyingCapability or ConstraintSatisfyingCapability` | capability with a reason to be in a plan |
| `ConstraintSatisfyingSkill EquivalentTo Skill and (hasCapability some ConstraintSatisfyingCapability)` | skill needed to meet a hard constraint |

**Modeling decisions.** The classes are defined, not asserted: a capability is justified because of the links it has, so adding a requirement or a hard constraint to the KG changes the classification without editing the capability. They are global, meaning a capability is justified if it meets some requirement or hard constraint of some request, so the competency-question queries keep the request in the query.

**Competency questions.** CQ 3 (Jessica, a focus question): the reasoner infers that the safety screening and evidence verification skills are ConstraintSatisfyingSkills, because their capabilities satisfy her two hard constraints. CQ 5 (Dante, a focus question): the query lists, for each task of his selected plan, the requirement or hard constraint that the required capability satisfies; that link is also why the reasoner classifies each of these capabilities as a JustifiedCapability.

**Changes in 4.0.** The CQ 5 sentence now says what the query returns.

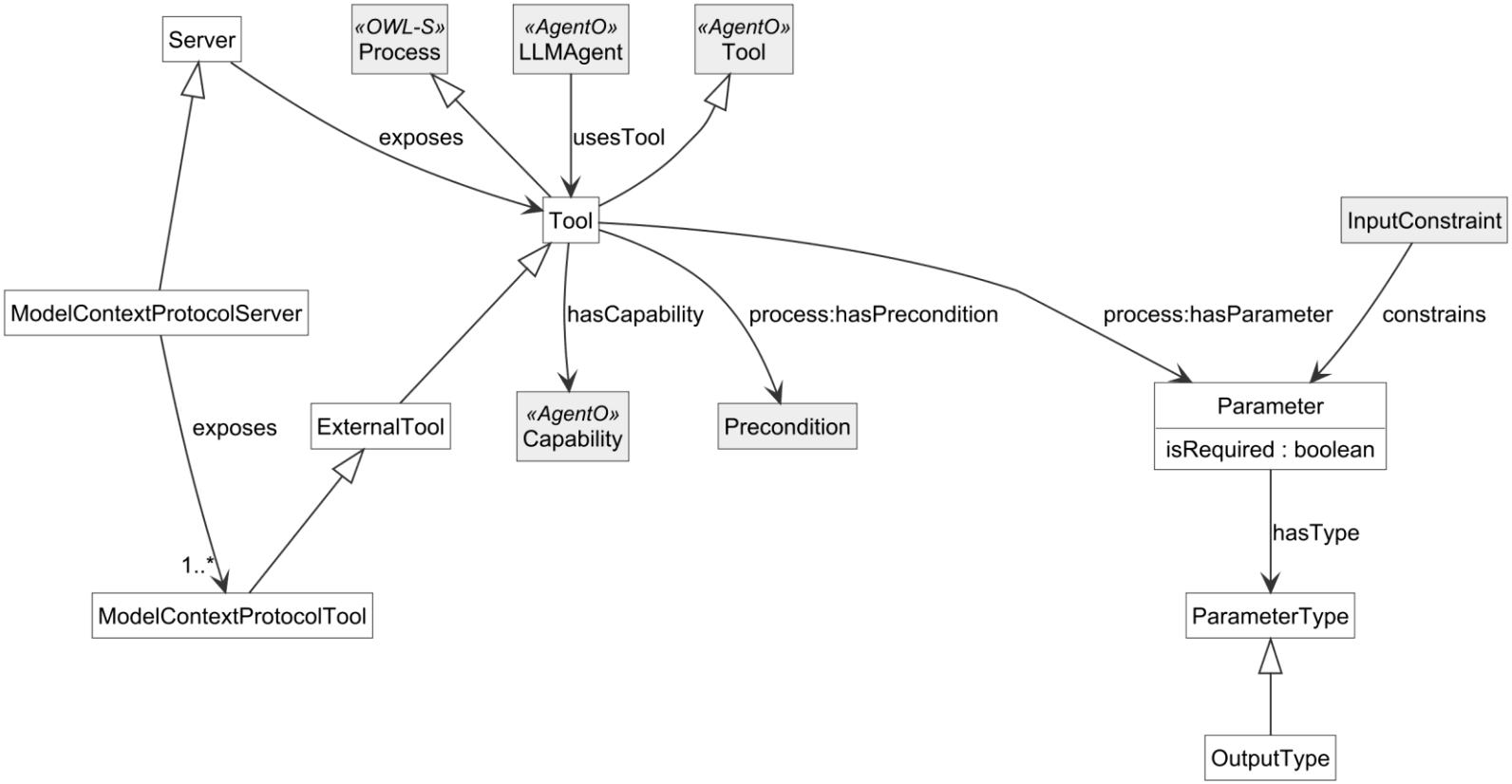



**Figure 4. Tools, servers and tool interfaces**

**Shows.** A Server exposes Tools. The PubMed, Supplement Advisor and OpenFDA servers are ModelContextProtocolServers [4] whose tools are ModelContextProtocolTools; iHerb-CLI's commands are ExternalTools reached through a command line. A Tool's interface is a set of Parameters, each with a ParameterType; OutputType types what a tool returns, isRequired marks required parameters, and an InputConstraint constrains a parameter's values. The LLMAgent uses tools. Each ModelContextProtocolServer exposes at least one ModelContextProtocolTool (1..*).

**Modeling decisions.** Interface facts (parameters, types, preconditions) are kept apart from run-time facts (Figure 7a), so the harness can check a plan before running it, as the third requirement in Section II of the use case asks. Tool is a subclass of agento:Tool and of OWL-S process:Process [8], so the AgentO and OWL-S properties apply to it unchanged.

**Competency questions.** CQ 9 checks required parameters (a RequiredInput fills a Parameter whose isRequired is true, Figure 8b) and preconditions before a call. The CQ 1 queries (step 1 and query 2) and the CQ 5 query name the MCP server that exposes each tool. Alternate flow 2 lists the available tools with their capabilities.

**Changes in 4.0.** Competency-question paragraph updated for the Assignment 10 queries.

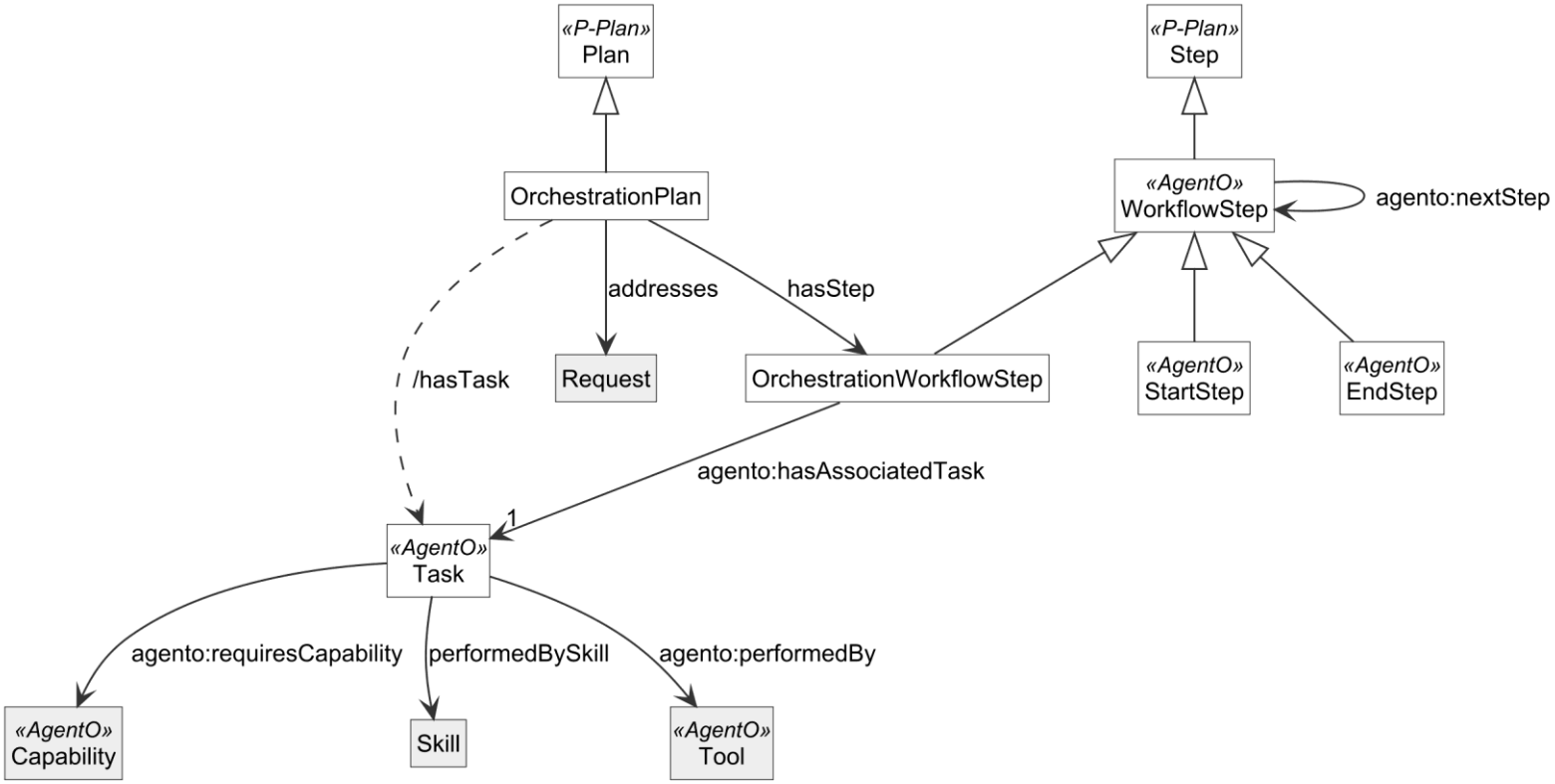



**Figure 5a. Plan structure**

**Shows.** An OrchestrationPlan is a P-Plan Plan [12]. It addresses one Request and has OrchestrationWorkflowSteps (hasStep), a kind of AgentO WorkflowStep; agento:nextStep orders the steps from a StartStep to an EndStep. Each OrchestrationWorkflowStep is associated with exactly one Task (multiplicity 1). A Task requires one or more Capabilities and is performed either by a Skill (performedBySkill) or by a Tool (agento:performedBy). /hasTask links a plan to the tasks of its steps.

**Modeling decisions.** hasTask is derived: hasStep o agento:hasAssociatedTask SubPropertyOf hasTask, so the shortcut cannot disagree with the steps. A task performed by a tool reuses agento:performedBy exactly as AgentO defines it, from Task to Tool. A task performed by a skill uses the new performedBySkill, named after AgentO's performedByAgent; it is not a subproperty of agento:performedBy, because that property's range would make every Skill a Tool.

**Competency questions.** CQ 1 query 2 (Ken, a focus question) lists the selected plan's steps (hasStep) with each step's task (agento:hasAssociatedTask), the skill that performs it (performedBySkill) and the tools the skill uses. The CQ 2, CQ 4 and CQ 9 queries start from the selected plan's steps and tasks in the same way. CQ 5 (Dante, a focus question) follows OrchestrationWorkflowStep → Task → agento:requiresCapability → Capability, and performedBySkill names the skill. /hasTask is the first link of the plan shortcuts planConsumes and planProduces (Figure 9e).

**Changes in 4.0.** Competency-question paragraph rewritten for the Assignment 10 queries (it was empty in the Assignment 9 copy).

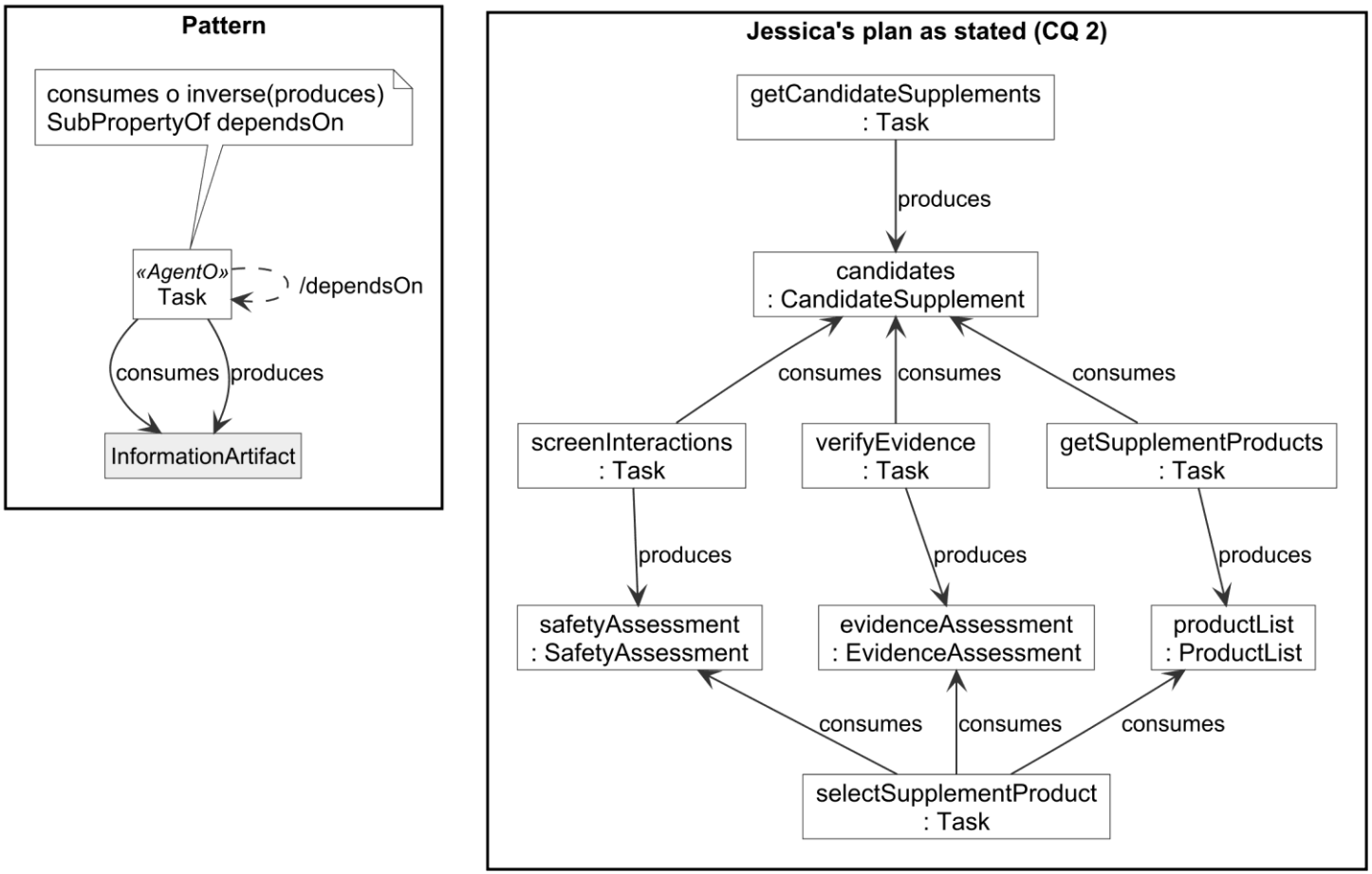



**Figure 5b. Task dependencies: the pattern and Jessica's plan as stated**

**Shows.** Left, the pattern: a Task consumes and produces InformationArtifacts, and dependsOn is derived from them. Right, Jessica's plan as the KG stores it (CQ 2): only consumes and produces are stated. Figure 5c shows the dependsOn links that the reasoner infers from it.

**Modeling decisions.** Dependencies are derived, not entered by hand. Task B dependsOn Task A when B consumes an artifact that A produces, written in Protégé as consumes o inverse(produces) SubPropertyOf dependsOn [14]. The KG therefore never stores a dependency that disagrees with the artifacts.

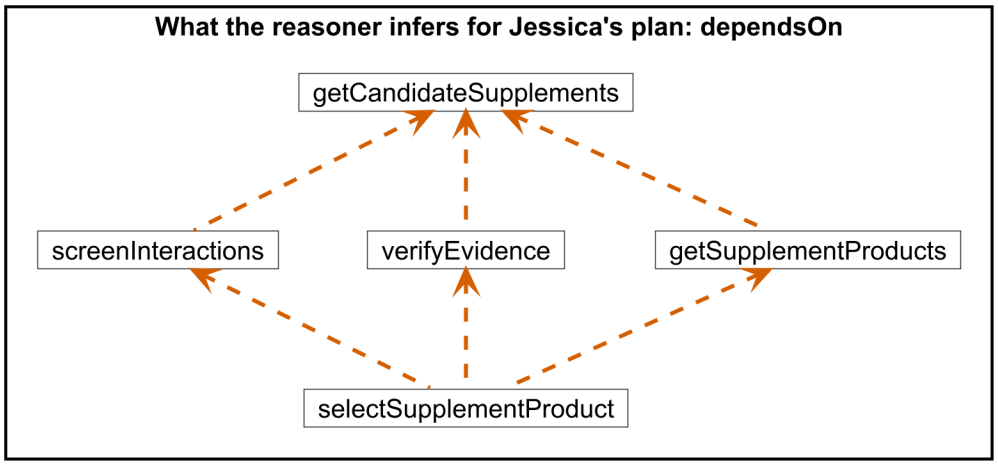



**Figure 5c. Task dependencies the reasoner infers for Jessica's plan**

**Shows.** The dependsOn links that Pellet infers from the stated links in Figure 5b, drawn as orange dashed arrows. The tasks are in the same order as in Figure 5b.

**Competency questions.** CQ 2 (Jessica): screenInteractions, verifyEvidence and getSupplementProducts each depend only on getCandidateSupplements, and no dependsOn links any two of them, so the three can run in parallel; selectSupplementProduct depends on all three and runs after them. The CQ 2 query keeps the step pairs whose tasks consume the same artifact, drops the pairs linked by dependsOn (MINUS, since the open-world assumption cannot infer that a link is missing), and adds the hard constraint each parallel step meets (stepSatisfiesHardConstraint, Figure 9c) with the preference setting behind it. CQ 4 (Dante) and CQ 8 read the order from the same kind of inferred links.

**Changes in 4.0.** The CQ 2 sentences now say what the query does.

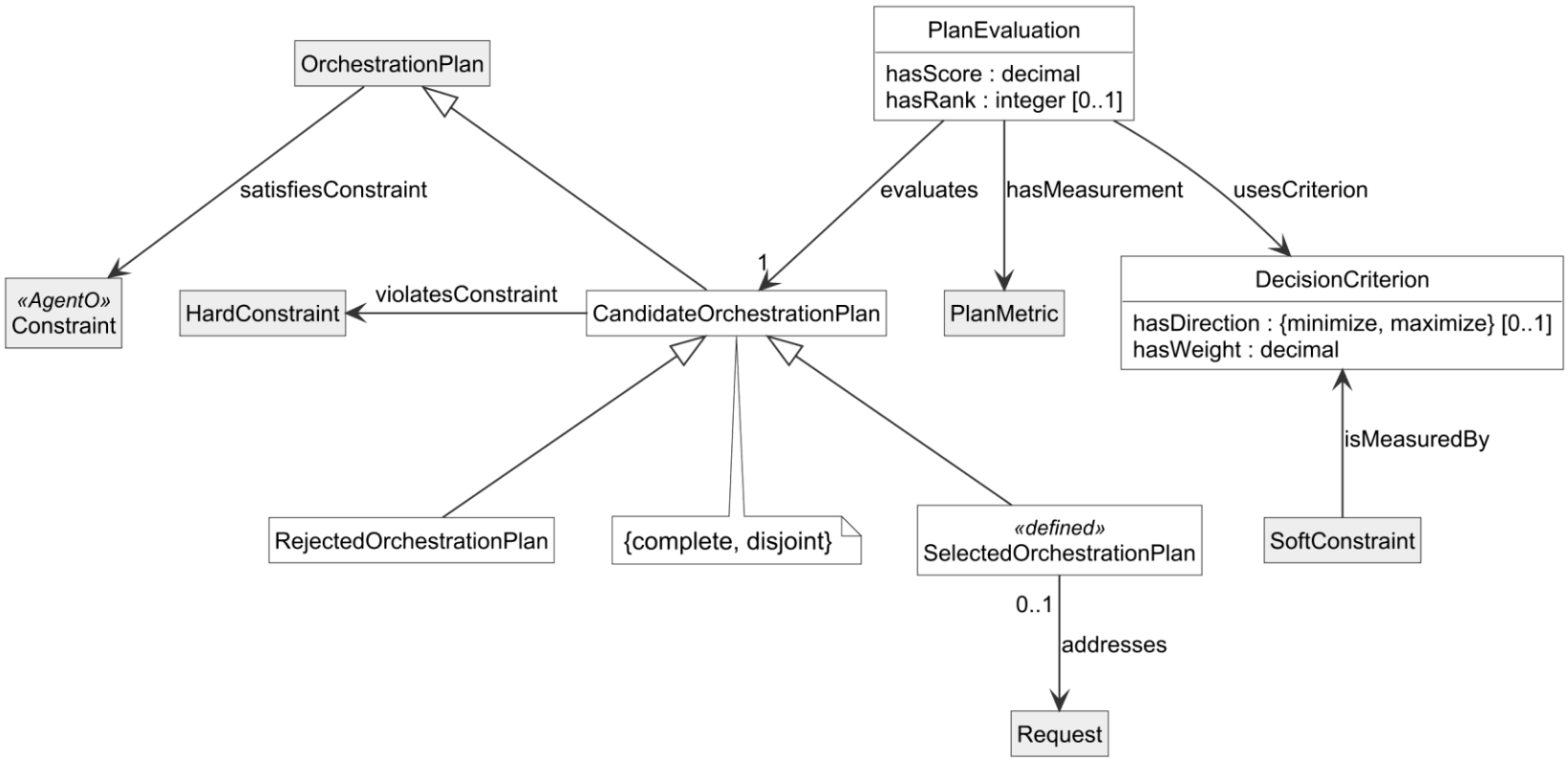



**Figure 6a. Plan evaluation and selection**

**Shows.** Candidate plans for a request are compared on DecisionCriteria. A PlanEvaluation evaluates exactly one CandidateOrchestrationPlan: it records the DecisionCriteria used, the PlanMetrics measured (Figure 6b), and the plan's score and rank, following the n-ary relation pattern [13]. A DecisionCriterion has a direction (minimize or maximize) and a weight taken from the priority of the SoftConstraint it measures. The axioms:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `CandidateOrchestrationPlan EquivalentTo SelectedOrchestrationPlan or RejectedOrchestrationPlan` | every candidate is selected or rejected |
| `SelectedOrchestrationPlan DisjointWith RejectedOrchestrationPlan` | never both |
| `SelectedOrchestrationPlan EquivalentTo CandidateOrchestrationPlan and (inverse evaluates some (hasRank value 1))` | the candidate ranked first is the selected plan |
| `CandidateOrchestrationPlan and (violatesConstraint some HardConstraint) SubClassOf RejectedOrchestrationPlan` | a candidate that violates a hard constraint is rejected |
| `Request SubClassOf inverse addresses max 1 SelectedOrchestrationPlan` | a request has at most one selected plan |
| `PlanEvaluation SubClassOf evaluates exactly 1 CandidateOrchestrationPlan` | an evaluation evaluates exactly one plan |
| `hasRank, hasDirection: Functional` | one rank per evaluation, one direction per criterion |

**Modeling decisions.** Selection runs in three steps, which is how conflicting criteria are handled. Hard constraints filter first: violatesConstraint links a candidate only to a HardConstraint, and the ontology classifies a candidate that violates one as a RejectedOrchestrationPlan. The remaining candidates are ranked by their weighted criterion scores, which the scoring step computes outside OWL; a soft constraint is never violated, it only lowers a plan's score and rank. Soft constraints that directly conflict (Figure 2) stop planning and go back to the user. The reasoner then classifies the plans: the candidate whose evaluation has rank 1 is a SelectedOrchestrationPlan (a defined class), and because a request has at most one selected plan and every candidate is selected or rejected, every other candidate for the same request is a RejectedOrchestrationPlan. This needs the plans to be stated as different individuals (owl:AllDifferent), since OWL does not assume that different names denote different plans.

**Competency questions.** CQ 1 (Ken, a focus question): on the example data the reasoner finds no violated hard constraint; plan 1's evaluation has score 2.00 and rank 1 (4 tool calls, $0.02 in tokens) and plan 2's has 0.00 and rank 2 (9 tool calls, $0.05), with Ken's two soft constraints at the same priority, so the reasoner classifies plan 1 as a SelectedOrchestrationPlan and plan 2 as a RejectedOrchestrationPlan. CQ 6 (Olivia) compares plans A and B the same way on evidence coverage and tool calls. CQ 7 follows each plan's evaluation (usesCriterion) to the criterion that measures each of Olivia's soft constraints (isMeasuredBy) and lists the hard constraints a plan violates (none for Olivia). CQ 4 (Dante) reads the criterion that measures his soft constraint "minimize execution time" and that his plan's evaluation uses.

**Changes in 4.0.** The CQ 4 and CQ 7 sentences now say what the queries read.

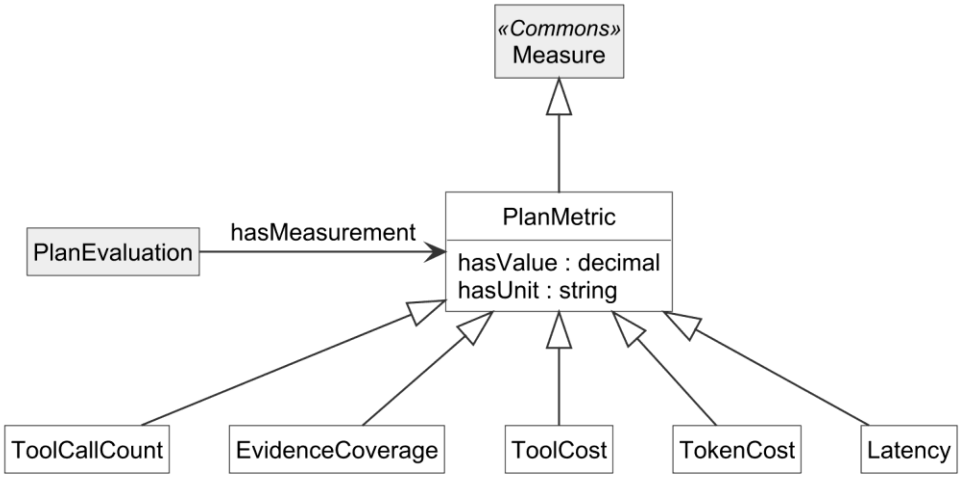



**Figure 6b. Plan metrics**

**Shows.** The metrics a PlanEvaluation records (hasMeasurement): ToolCallCount, EvidenceCoverage, ToolCost, TokenCost and Latency, each a PlanMetric with a value and a unit. PlanMetric is a Measure in the sense of the OMG Commons library [18].

**Competency questions.** CQ 1 (Ken) compares the plans on ToolCallCount and TokenCost, and CQ 6 and CQ 7 (Olivia) on EvidenceCoverage and ToolCallCount. The test data have no Latency measurement for Dante's plan, so CQ 4 reads only the criterion (Figure 6a).

**Changes in 4.0.** Competency-question paragraph updated for the Assignment 10 queries.

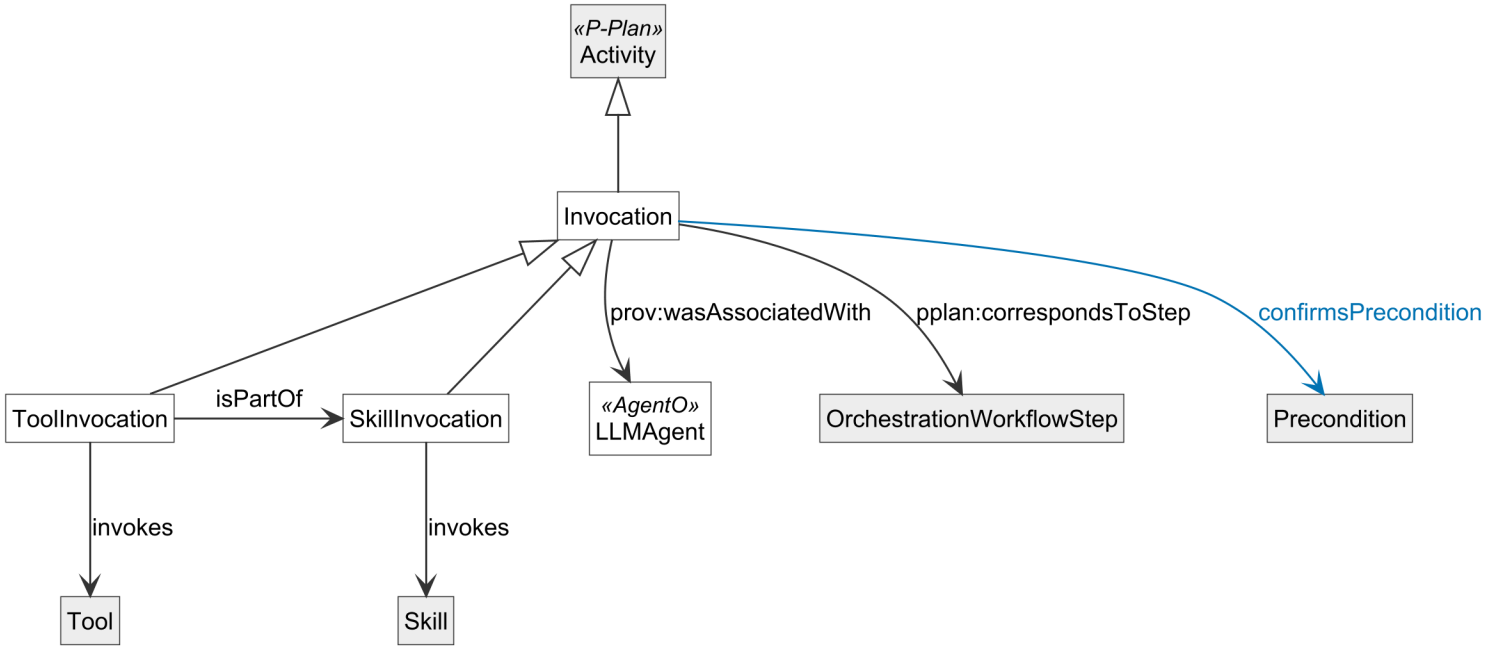



**Figure 7a. Invocations**

**Shows.** The run-time side of a plan, following P-Plan [12] and PROV-O [9]. An Invocation is a P-Plan Activity that corresponds to one OrchestrationWorkflowStep and is associated with the LLMAgent that ran it. A ToolInvocation invokes a Tool and is part of the SkillInvocation that made the call; the three OpenFDA calls in Jessica's plan, for example, are part of one Screen interactions invocation. An Invocation that checked a Precondition confirms it (confirmsPrecondition).

**Modeling decisions.** The plan (Figure 5a) and its execution are kept separate, so the KG can compare what was planned with what ran; reusing PROV-O and P-Plan instead of new terms keeps the record readable by existing provenance tools.

**Competency questions.** CQ 9 (Jessica): before the evidence verification skill runs, the query checks each of the skill's Preconditions (hasPrecondition, Figure 3a) for an Invocation that confirms it and that her ExecutionTrace records (Figure 7b). In the test data, a PubMed reachability check confirms the skill's precondition.

**Changes in 4.0.** New: confirmsPrecondition, added to the ontology for CQ 9; Precondition (Figures 2 and 3a) is drawn gray. The competency-question paragraph now names CQ 9, the query that reads these records.

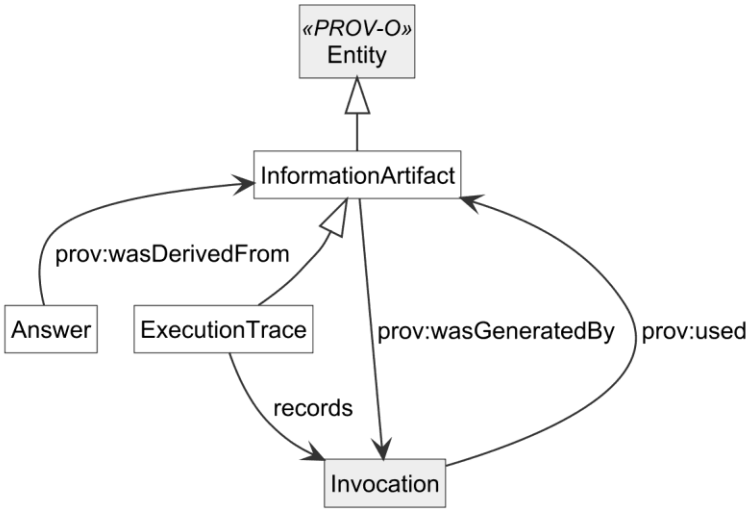



**Figure 7b. Provenance of information artifacts**

**Shows.** An InformationArtifact is a PROV-O Entity. An Invocation uses artifacts and generates new ones; the Answer is derived from the artifacts it summarizes; an ExecutionTrace is itself an information artifact and records the invocations, as basic flow step 11 requires.

**Competency questions.** CQ 9 checks which artifacts have been generated so far: a required input counts as generated when an invocation that the ExecutionTrace records generated it (prov:wasGeneratedBy).

**Changes in 4.0.** Competency-question paragraph updated for the CQ 9 query.

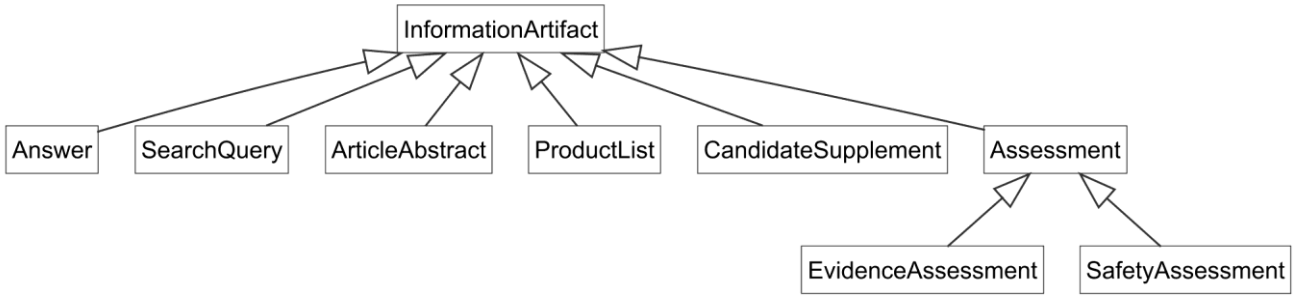



**Figure 8a. Kinds of information artifacts**

**Shows.** The kinds of artifacts that the use case flows pass between steps. A SearchQuery feeds the search tools; an ArticleAbstract is what PubMed download_articles_batch returns; a CandidateSupplement comes from the Supplement Advisor and a ProductList from iHerb-CLI; the Answer is the final response (Figure 7b). EvidenceAssessment and SafetyAssessment are Assessments of a CandidateSupplement.

**Modeling decisions.** Kinds are shown here and roles in Figure 8b. Assessment groups the two artifacts that judge a candidate supplement. The link from an assessment to its supplement (assessesSupplement) is not in the ontology.

**Competency questions.** The CQ 2 and CQ 4 queries read these kinds as what the tasks consume and produce: Jessica's candidate supplements (CQ 2); Dante's two CandidateSupplements, the EvidenceAssessment and SafetyAssessment of each, and his Answer (CQ 4). CQ 9 checks that Jessica's SearchQuery has been generated.

**Changes in 4.0.** Figure 8c is gone, so the modeling note no longer points to it; competency-question paragraph updated for the Assignment 10 queries.

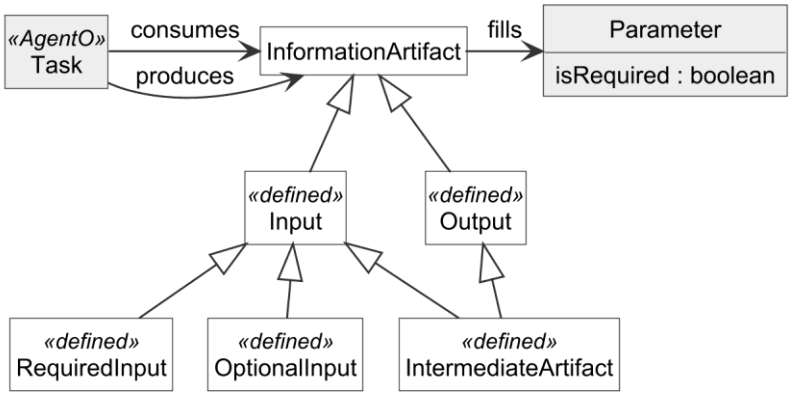



**Figure 8b. Roles of information artifacts**

**Shows.** Input, Output, RequiredInput, OptionalInput and IntermediateArtifact are roles that an artifact plays in a plan, not kinds, following the party and role pattern from class. They are «defined» classes, so the reasoner assigns them from consumes, produces and fills:

| Class | Definition (Manchester syntax) |
|---|---|
| Input | `InformationArtifact and (inverse consumes some Task)` |
| Output | `InformationArtifact and (inverse produces some Task)` |
| IntermediateArtifact | `Input and Output` |
| RequiredInput | `Input and (fills some (Parameter and (isRequired value true)))` |
| OptionalInput | `Input and (fills some (Parameter and (isRequired value false)))` |

**Modeling decisions.** Because the roles are defined, an artifact's role follows the plan instead of being entered by hand. In Jessica's plan the reasoner classifies the candidate supplements as an Output, an Input and an IntermediateArtifact.

**Competency questions.** CQ 9: before verifyEvidence runs, the harness checks that each RequiredInput it consumes has been generated (prov:wasGeneratedBy, Figure 7b). The system writes these records itself, so this check can use a closed-world query. In the test data her SearchQuery fills the required PubMed query parameter and has been generated, but the candidate supplements have not, so the answer names the skill that must run first, the personalized supplement recommendation skill.

**Changes in 4.0.** Last sentence added: the answer the CQ 9 query returns.

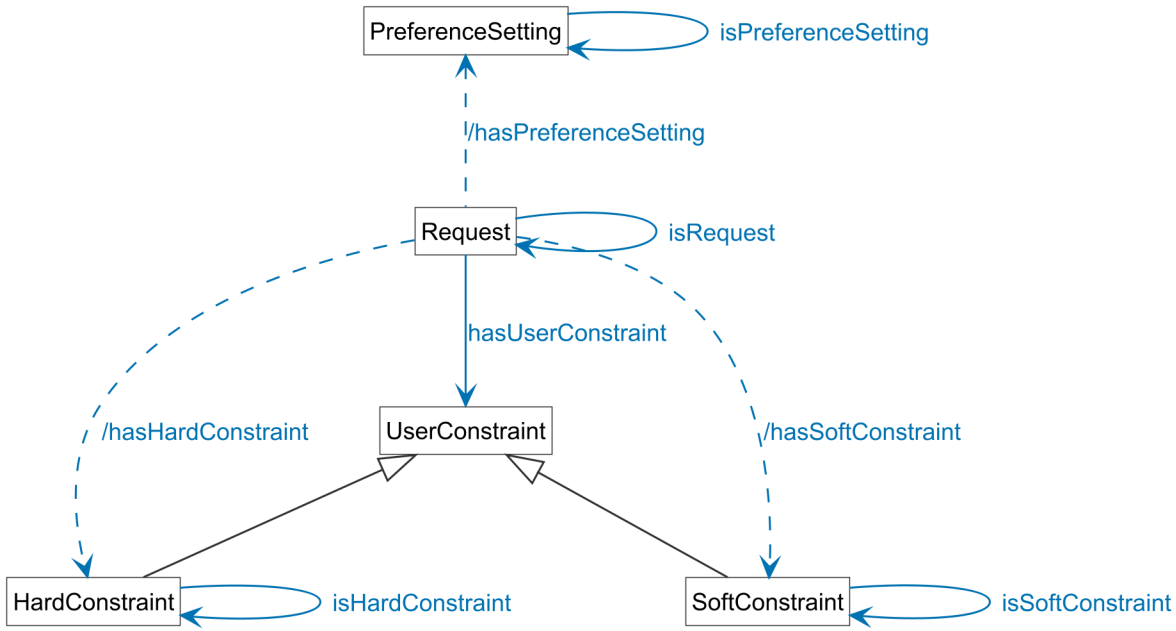



**Figure 9a. Shortcuts from a request to its preference settings and constraints**

**Shows.** Four properties link a Request directly to what the queries ask about. /hasPreferenceSetting links a request to the preference settings of the person who made it. hasUserConstraint links a request to its user constraints, and its two subproperties /hasHardConstraint and /hasSoftConstraint to the hard and the soft ones. The loops are the helper properties the chains use. The axioms:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `isRequest o prov:wasAttributedTo o inverse(prov:wasAttributedTo) o isPreferenceSetting SubPropertyOf hasPreferenceSetting` | settings of the person who made the request |
| `isRequest o inverse(prov:wasDerivedFrom) o isHardConstraint SubPropertyOf hasHardConstraint` | hard constraint derived from the request |
| `hasPreferenceSetting o inverse(prov:wasDerivedFrom) o isHardConstraint SubPropertyOf hasHardConstraint` | hard constraint derived from its settings |
| `isRequest o inverse(prov:wasDerivedFrom) o isSoftConstraint SubPropertyOf hasSoftConstraint` | soft constraint derived from the request |
| `hasPreferenceSetting o inverse(prov:wasDerivedFrom) o isSoftConstraint SubPropertyOf hasSoftConstraint` | soft constraint derived from its settings |
| `hasHardConstraint, hasSoftConstraint SubPropertyOf hasUserConstraint` | both kinds are user constraints |
| `Request EquivalentTo isRequest Self; likewise PreferenceSetting, HardConstraint, SoftConstraint` | each member is linked to itself |

**Modeling decisions.** In the KG a constraint points back to where it came from (prov:wasDerivedFrom, Figure 2), and a request and a preference setting are linked only through the prov:Person they are attributed to. Each shortcut is defined by property chains over those stated links, so the reasoner infers it and it cannot disagree with them. A chain cannot name a class, so it passes through a helper loop: a chain that ends in isHardConstraint reaches only members of HardConstraint, because HardConstraint EquivalentTo isHardConstraint Self. hasHardConstraint and hasSoftConstraint each have two chains, for a constraint derived from the request itself and for one derived from the request's preference settings.

**Competency questions.** The CQ 1 (step 1 and query 1), CQ 2 and CQ 7 queries read a request's hard constraints, and the CQ 4 and CQ 7 queries its soft constraints, through these shortcuts. Without them each query had to write out the path through prov:wasAttributedTo and prov:wasDerivedFrom, as the CQ 3 query still does, because Snap SPARQL has no property paths.

**Changes in 4.0.** New figure: properties added to the ontology for Assignment 10.

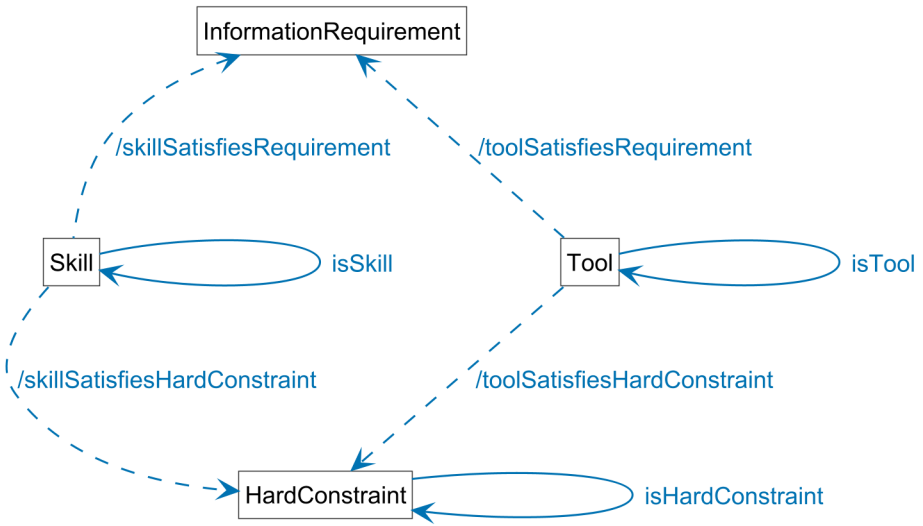



**Figure 9b. Shortcuts from skills and tools to requirements and hard constraints**

**Shows.** Four shortcuts link a Skill or a Tool directly to the InformationRequirements and HardConstraints that its capabilities satisfy (Figure 3b). The axioms:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `isSkill o hasCapability o satisfiesRequirement SubPropertyOf skillSatisfiesRequirement` | a capability of the skill meets the requirement |
| `isSkill o hasCapability o satisfiesConstraint o isHardConstraint SubPropertyOf skillSatisfiesHardConstraint` | a capability of the skill meets the hard constraint |
| `isTool o hasCapability o satisfiesRequirement SubPropertyOf toolSatisfiesRequirement` | a capability of the tool meets the requirement |
| `isTool o hasCapability o satisfiesConstraint o isHardConstraint SubPropertyOf toolSatisfiesHardConstraint` | a capability of the tool meets the hard constraint |
| `Skill EquivalentTo isSkill Self; Tool EquivalentTo isTool Self` | each member is linked to itself |

**Modeling decisions.** Skills and tools both have capabilities (hasCapability), so the isSkill and isTool loops at the start of the chains keep the skill shortcuts and the tool shortcuts apart. satisfiesConstraint ranges over all constraints, so the chains for hard constraints end in isHardConstraint.

**Competency questions.** CQ 1 step 1 (Ken, a focus question) uses all four: for each requirement his request expresses and each hard constraint it has (Figure 9a), the skills and tools that meet it, with the MCP server that exposes each tool (Figure 4).

**Changes in 4.0.** New figure: properties added to the ontology for Assignment 10.

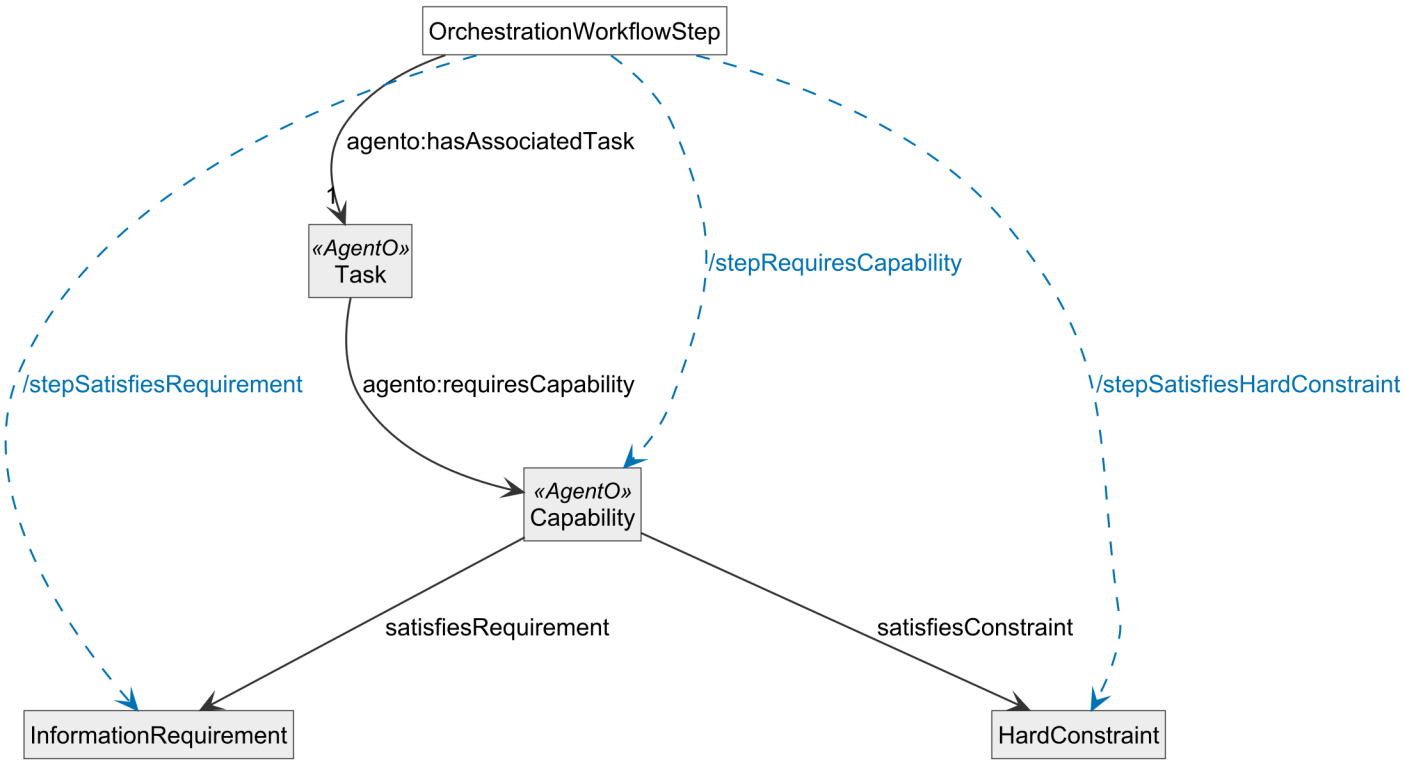



**Figure 9c. Shortcuts from workflow steps**

**Shows.** Three shortcuts link an OrchestrationWorkflowStep to the Capability its task requires and to the InformationRequirements and HardConstraints that capability satisfies. The gray classes and black arrows are the stated links the chains follow. The axioms:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `agento:hasAssociatedTask o agento:requiresCapability SubPropertyOf stepRequiresCapability` | the capability the step's task requires |
| `stepRequiresCapability o satisfiesRequirement SubPropertyOf stepSatisfiesRequirement` | requirement that capability meets |
| `stepRequiresCapability o satisfiesConstraint o isHardConstraint SubPropertyOf stepSatisfiesHardConstraint` | hard constraint that capability meets |

**Modeling decisions.** The two longer chains start with stepRequiresCapability, so the step-to-capability link is defined once. The requirement or hard constraint a step meets is what justifies the step in its plan.

**Competency questions.** CQ 1 query 2 (Ken, a focus question) lists the steps of the selected plan with the requirement or hard constraint each meets. The CQ 2 query (Jessica) uses stepSatisfiesHardConstraint for the hard constraint each parallel step meets.

**Changes in 4.0.** New figure: properties added to the ontology for Assignment 10.

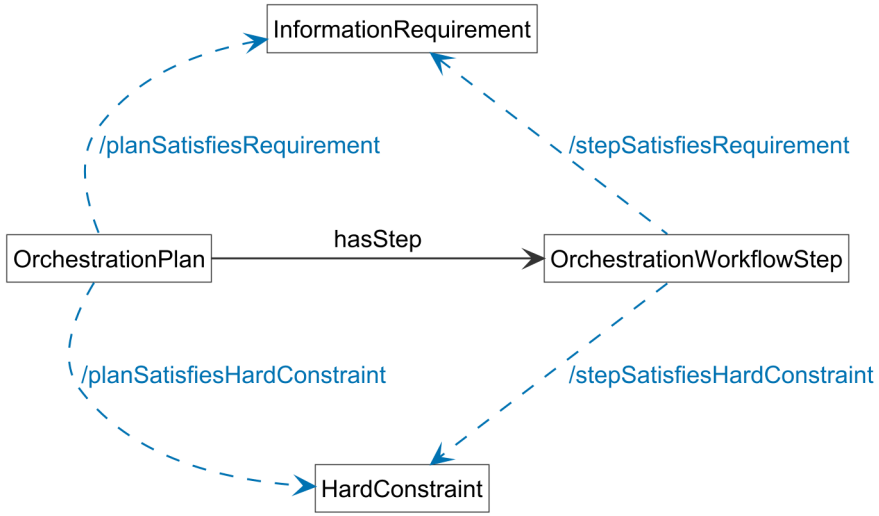



**Figure 9d. Shortcuts from plans to requirements and hard constraints**

**Shows.** A plan satisfies a requirement or a hard constraint when one of its steps does (Figure 9c). The axioms:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `hasStep o stepSatisfiesRequirement SubPropertyOf planSatisfiesRequirement` | a step of the plan meets the requirement |
| `hasStep o stepSatisfiesHardConstraint SubPropertyOf planSatisfiesHardConstraint` | a step of the plan meets the hard constraint |

**Modeling decisions.** The shortcuts say that some step meets a requirement. That a plan meets every requirement and every hard constraint of the request cannot be inferred under the open-world assumption, so CQ 1 query 1 checks it in SPARQL with MINUS.

**Competency questions.** CQ 1 query 1 (Ken, a focus question) keeps the candidate plans that meet every requirement his request expresses and every hard constraint it has.

**Changes in 4.0.** New figure: properties added to the ontology for Assignment 10.

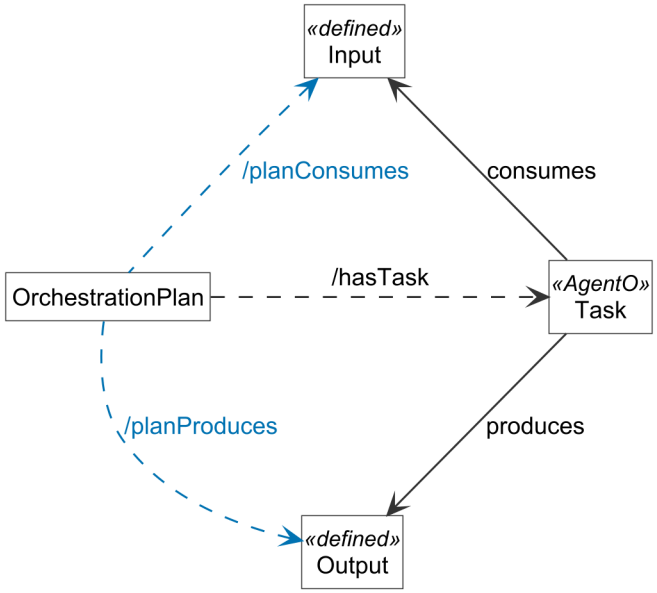



**Figure 9e. Shortcuts from plans to the artifacts their tasks consume and produce**

**Shows.** /planConsumes and /planProduces link a plan to every artifact that one of its tasks consumes or produces, through the derived /hasTask of Figure 5a. The targets are drawn as the defined roles Input and Output of Figure 8b, which is what the reasoner classifies them as; the declared range of both properties is InformationArtifact. The axioms:

| Axiom (Manchester syntax) | Meaning |
|---|---|
| `hasTask o consumes SubPropertyOf planConsumes` | artifact a task of the plan consumes |
| `hasTask o produces SubPropertyOf planProduces` | artifact a task of the plan produces |

**Competency questions.** CQ 1 query 1 (Ken, a focus question) uses both to keep only the candidate plans that produce every artifact they consume, so that no step waits for an input that no step makes.

**Changes in 4.0.** New figure: properties added to the ontology for Assignment 10.

## How the model answers the competency questions

Bold rows are the three focus questions that scope the ontology. Every row has a SPARQL query in OE_10_TeamAgenticCapabilities_Queries.txt; on November 6, 2026 each returned its expected answer in Snap SPARQL 6.0.0 with Pellet and with HermiT, on the 162 test individuals of OE_10_TeamAgenticCapabilities_Individuals.rdf, which follow the usage scenarios (see the use case, Section VII). Blue rows changed in Version 4.0.

| CQ | Scenario | Figures | Path through the model (as the query follows it) |
|---|---|---|---|
| **1. What orchestration should the agent follow? (focus)** | **Ken** | **2, 3b, 4, 5a, 6a, 6b, 9a–9e** | **Step 1: Request → expresses → InformationRequirement, and /hasHardConstraint → HardConstraint; /skillSatisfies… and /toolSatisfies… name the Skills and Tools that meet each; usesTool and exposes give tools and MCP servers. Query 1: CandidateOrchestrationPlans whose /planSatisfiesRequirement and /planSatisfiesHardConstraint cover the request, that violate no HardConstraint and produce (/planProduces) every artifact they consume (/planConsumes); PlanEvaluations: ToolCallCount and TokenCost, hasScore 2.00 vs 0.00, hasRank 1 vs 2 → inferred SelectedOrchestrationPlan and RejectedOrchestrationPlan. Query 2: selected plan → hasStep → steps with /stepSatisfiesRequirement or /stepSatisfiesHardConstraint, each task's skill (performedBySkill) and tools** |
| 2. Which steps can run in parallel? | Jessica | 5a, 5b, 5c, 9a, 9c | Selected plan → hasStep → step → agento:hasAssociatedTask → Task; tasks consume and produce artifacts → derived dependsOn; steps whose tasks share an input and are not linked by dependsOn (MINUS) run in parallel: screenInteractions, verifyEvidence and getSupplementProducts; /stepSatisfiesHardConstraint gives the hard constraint each meets, from her preference settings (/hasHardConstraint) |
| **3. What skills are required? (focus)** | **Jessica** | **2, 3b, 3c** | **Request → expresses → InformationRequirement ← satisfiesRequirement ← Capability; Request → prov:wasAttributedTo → Person ← prov:wasAttributedTo ← PreferenceSetting ← prov:wasDerivedFrom ← HardConstraint ← satisfiesConstraint ← Capability; Skill → hasCapability → those capabilities; the skills whose capability meets a hard constraint are inferred ConstraintSatisfyingSkills** |
| 4. Why were capabilities invoked in this order? | Dante | 2, 5c, 6a, 8a, 9a | Selected plan → hasStep → steps → tasks; the tasks consume and produce his two CandidateSupplements and four assessments → derived dependsOn: steps 1a and 1b, and 2a and 2b, share an input and do not depend on each other, so each pair runs together, which /hasSoftConstraint "minimize execution time" → isMeasuredBy → criterion ← usesCriterion ← his PlanEvaluation favors; step 3 consumes the four assessments, so it runs last |
| **5. Why was each capability chosen? (focus)** | **Dante** | **3b, 3c, 4, 5a** | **Selected plan → hasStep → OrchestrationWorkflowStep → agento:hasAssociatedTask → Task → agento:requiresCapability → Capability → satisfiesRequirement (a requirement the request expresses) or satisfiesConstraint (the safety-screening HardConstraint derived from his PreferenceSetting); performedBySkill, usesTool and exposes name the skill, tools and server. A second query gives the same rows without a reasoner, reading rank 1 instead of the inferred class** |
| 6. Why was plan B selected over plan A? | Olivia | 6a, 6b | PlanEvaluations evaluate plans A and B → hasMeasurement: EvidenceCoverage 1.00 vs 0.70, ToolCallCount 8 vs 3 → hasScore 1.00 vs 1.70, hasRank 2 vs 1 → B inferred SelectedOrchestrationPlan, A RejectedOrchestrationPlan |
| 7. Which constraints were violated by a | Olivia | 2, 6a, 6b, 9a | Request → /hasHardConstraint → HardConstraint ← violatesConstraint ← plan (none for |

| | | | |
|---|---|---|---|
| candidate plan? | | | Olivia); Request → /hasSoftConstraint → SoftConstraint → isMeasuredBy → DecisionCriterion ← usesCriterion ← each plan's PlanEvaluation, with hasPriority and the plan's rank and score: plan A ranks second on tool calls (8 vs 3) despite full evidence coverage |
| 8. What skills depend on other skills? | — | 5a, 5c | Skill ← performedBySkill ← Task; tasks' consumes and produces → derived dependsOn → seven skill dependencies, for example evidence synthesis depends on evidence verification and on safety screening |
| 9. Are the prerequisites satisfied before a skill runs? | Jessica | 3a, 4, 7a, 7b, 8b | Selected plan → step whose task the evidence verification skill performs; Skill → hasPrecondition → Precondition ← confirmsPrecondition ← Invocation that the ExecutionTrace records; Task → consumes → RequiredInput (inferred) ← prov:wasGeneratedBy ← recorded Invocation; a missing input leads to the task that produces it and its skill: the candidate supplements are missing, so the recommendation skill must run first |

## References

Numbers [1]–[14] follow Section IX of the use case; [18] was added in Version 2.0.


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